Assignment No: 5

Title: Apply CSS Positions & other properties.

Aim: To study and apply the CSS position property in all of its values - static, relative, absolute, fixed and sticky - together with supporting properties such as z-index, overflow, opacity, box-shadow, border-radius, transition and transform.

Objectives:

1. To understand the CSS box model and the normal document flow.
2. To apply the five values of the position property and observe their effect.
3. To control the stacking order of overlapping elements using z-index.
4. To build an overlay that covers its parent element exactly.
5. To apply visual properties such as box-shadow, border-radius, opacity and backdrop-filter.
6. To create smooth state changes using transition and transform.

Hardware Used:

- Computer or Laptop
- Keyboard
- Mouse
- Minimum 4 GB RAM

Software Used:

- Visual Studio Code (VS Code)
- Web Browser (Google Chrome, Microsoft Edge, or Firefox)
- Operating System: Windows / Linux / macOS

Theory:

1. The CSS Box Model

Every HTML element is rendered as a rectangular box made of four layers: the content, the padding around it, the border around that, and the margin outside the border. The position property decides where that box is placed.

2. The position Property

Value	Behaviour
static	Default. The element stays in the normal flow; top/left are ignored.
relative	Stays in the flow but is offset from its normal place; becomes a containing block for absolutely positioned children.
absolute	Removed from the flow and positioned against the nearest positioned ancestor.

fixed	Removed from the flow and positioned against the viewport; does not move when the page scrolls.
sticky	Behaves as relative until a scroll threshold is crossed, then behaves as fixed within its parent.

3. Offset Properties and z-index

top, right, bottom and left move a positioned element. Setting all four to 0 on an absolutely positioned element makes it cover its positioned parent exactly. z-index controls the stacking order of overlapping positioned elements - a higher value paints on top. z-index only applies to elements whose position is not static.

Full-cover overlay pattern:

```
.parent  { position: relative; }
.overlay { position: absolute; top: 0; left: 0; right: 0; bottom: 0;
           z-index: 20; }
```

4. Other Visual Properties

Property	Purpose
overflow	Controls content that exceeds the box (visible, hidden, scroll, auto)
opacity	Transparency of the element from 0 to 1
box-shadow	Draws a shadow: offset-x offset-y blur spread color
border-radius	Rounds the corners of the box
backdrop-filter	Applies a filter such as blur() to whatever is behind the box
transition	Animates a property change over a duration
transform	Moves, rotates or scales the element without affecting the flow

5. Sticky Navigation

A navigation bar declared with position: sticky and top: 0 scrolls normally until it reaches the top of the viewport and then stays pinned there. Unlike position: fixed it reserves its space in the flow, so the page content does not jump when the bar becomes pinned.

Course Project Implementation:

Course-Project Mapping: Fully available in the course project. Submit src/components/Navbar.jsx (sticky positioning with z-index and backdrop-filter), src/components/UploadSection.jsx (a relative container with a full-cover absolute overlay, stacking order and transform animation) and src/index.css (box-shadow, border-radius, overflow and custom scrollbar styling).

Positioning concept	File	Evidence
position: sticky	src/components/Navbar.jsx	sticky top-0 z-50 backdrop-blur-lg

position: absolute (viewport corner)	src/App.jsx	absolute top-4 right-4 z-[100]
relative parent + absolute overlay	src/components/UploadSection.jsx	group relative wrapper with an inset-0 loading overlay at zIndex 20
z-index stacking	src/components/UploadSection.jsx	image at z-0, gradient at z-10, overlay at z-20
overflow / border-radius / box-shadow	src/index.css	glass-card, custom ::-webkit-scrollbar rules
transform + transition	src/components/Navbar.jsx	transition-transform duration-500 group-hover:scale-105

Program:

Project file - src/components/Navbar.jsx (position: sticky)

```
<nav
  className="sticky top-0 z-50 backdrop-blur-lg border-b shadow-2xl transition-colors"
  style={{ backgroundColor: theme.headerBg, borderColor: theme.border }}
>
  {/* sticky  -> scrolls with the page, then pins at top: 0
      z-50    -> paints above the page content
      backdrop-blur-lg -> backdrop-filter: blur(16px) on what is behind it */}

  <img
    src={logo}
    alt="LipiSutra"
    className="h-40 w-auto object-contain relative z-10
               transition-transform duration-500 group-hover:scale-105"
  />
</nav>
```

Project file - src/App.jsx (position: absolute against the viewport corner)

```
<header className="absolute top-4 right-4 z-[100]">
  <button
    onClick={toggleTheme}
    className="px-4 py-2 rounded-full font-semibold shadow-md flex items-center gap-2"
    style={{
      backgroundColor: themeMode === 'LIGHT' ? theme.accent : theme.surface,
      border: `1.5px solid ${theme.border}`,
      transition: "all 0.3s"          /* smooth colour change */
    }}
  >
    {themeMode === "DARK" ? "Dark" : "Light"}
  </button>
</header>
```

Project file - src/components/UploadSection.jsx (relative parent + absolute overlay)

```
{/* The parent establishes the containing block */}
<div className="p-4 rounded-2xl border shadow-2xl w-full flex justify-center
                overflow-hidden relative group">

  {/* layer 0 : the artifact image */}
  <img src={image} alt="Historical Document"
       className="w-full h-auto rounded-xl object-contain max-h-[600px] relative z-
0" />
```

```
  {/* layer 10 : a hover gradient, click-through via pointer-events-none */}
  <div className="absolute inset-0 bg-gradient-to-t from-museum-900/80 to-transparent
                  opacity-0 group-hover:opacity-100 transition-opacity z-10
                  pointer-events-none"></div>

  {/* layer 20 : the loading overlay, covering the parent exactly */}
  {loading && (
    <div style={{
      position: 'absolute',
      top: 0, left: 0, right: 0, bottom: 0,     /* full cover */
      display: 'flex',
      flexDirection: 'column',
      alignItems: 'center',
      justifyContent: 'center',
      zIndex: 20,                               /* above image and gradient */
      backgroundColor: `${theme.bg}D9`,         /* D9 = 85% alpha */
      backdropFilter: 'blur(4px)',
      borderRadius: '0.75rem'
    }}>

      <style>{`
        @keyframes pulse-spin {
          0%   { transform: rotate(0deg)   scale(0.9);
                 box-shadow: 0 0 0  0px ${theme.accent}66; }
          50%  { transform: rotate(180deg) scale(1.1);
                 box-shadow: 0 0 0 20px ${theme.accent}00; }
          100% { transform: rotate(360deg) scale(0.9);
                 box-shadow: 0 0 0  0px ${theme.accent}00; }
        }
      `}</style>

      {/* the spinner : transform + box-shadow + border-radius */}
      <div style={{
        width: '64px', height: '64px',
        border: `4px solid ${theme.accent}33`,
        borderTopColor: theme.accent,
        borderRadius: '50%',
        animation: 'pulse-spin 1.5s cubic-bezier(0.4, 0, 0.2, 1) infinite'
      }}></div>

      {/* the progress bar : relative track, absolute fill */}
      <div style={{
        width: '75%', maxWidth: '320px', height: '8px',
        backgroundColor: theme.surface,
        borderRadius: '9999px',
        overflow: 'hidden',                     /* clips the fill */
        position: 'relative',
        boxShadow: 'inset 0 1px 3px rgba(0,0,0,0.5)'
      }}>
        <div style={{
          position: 'absolute', top: 0, left: 0,
          width: `${progress}%`, height: '100%',
          backgroundColor: theme.accent,
          transition: 'width 0.3s ease-out',    /* animated growth */
          boxShadow: `0 0 10px ${theme.accent}99`,
          borderRadius: '9999px'
        }}></div>
      </div>

    </div>
  )}
</div>
```

Project file - src/index.css (overflow, radius, shadow, scrollbar)

```
/* translucent card with a blurred backdrop */
.glass-card {
  background: rgba(26, 18, 11, 0.6);
```

```
  backdrop-filter: blur(8px);
  border: 1px solid rgba(201, 168, 76, 0.15);
}
.glass-card:hover {
  border-color: rgba(201, 168, 76, 0.3);
  background: rgba(26, 18, 11, 0.7);
}

body {
  background-attachment: fixed;   /* background does not scroll with content */
  min-height: 100vh;
}

/* custom scrollbar : overflow styling */
::-webkit-scrollbar       { width: 8px; }
::-webkit-scrollbar-track { background: var(--color-museum-900); }
::-webkit-scrollbar-thumb {
  background: var(--color-gold-900);
  border-radius: 4px;
  border: 1px solid var(--color-gold-600);
}
::-webkit-scrollbar-thumb:hover { background: var(--color-gold-600); }
```

Output:

- The navigation bar scrolls with the page and then pins itself to the top of the viewport, with the page content visible and blurred behind its translucent background.
- The theme toggle button stays anchored to the top-right corner above every other element because of z-[100].
- While a manuscript is being decoded, the loading overlay covers the artifact image exactly, dims and blurs it, and shows a spinner that rotates and scales via transform.
- The progress fill is clipped by its rounded track because the track sets overflow: hidden, and its width grows smoothly thanks to transition: width 0.3s ease-out.
- Hovering the image fades in the bottom gradient without blocking the click, because the gradient layer sets pointer-events: none.

Conclusion:

In this assignment, we applied the CSS position property in all five of its values and combined it with z-index, overflow, opacity, box-shadow, border-radius, backdrop-filter, transition and transform. The LipiSutra project demonstrates each of these in production code: a sticky navigation bar, an absolutely positioned theme toggle, and a three-layer stacking context in which an absolute overlay covers the artifact image while it is being decoded.

References:

7. Mozilla Developer Network (MDN) Web Docs - CSS Basics, https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Styling_basics
8. W3Schools - CSS Tutorial, https://www.w3schools.com/css/
9. CSS Specifications - W3C, https://www.w3.org/Style/CSS/
10. MDN Web Docs - position, https://developer.mozilla.org/en-US/docs/Web/CSS/position
11. MDN Web Docs - Stacking context, https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_positioned_layout/Stacking_context